



Daffodil
International
University

SRS

Smart Food Waste Reduction & Redistribution System

Course Title: Software Engineering Design Capstone Project

Course Code: SE-331

Course Teacher: Rahat Uddin Azad

Group Information:

Group Name: Social Warrior

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1. Introduction

1.1. Problem Statement:

Everyday a large amount of food is are being wasted in hotels, households and events because there is a lack of connection between food providers (from where foods are being wasted) and the needy people. The current processes rely on manual contact- phone calls or social media posts, which are slow, decentralized, difficult to track and not that much effective most of the time as the latency results in food expiry before redistribution.

1.2. Purpose:

The primary objective is to develop a centralized web platform that automates the coordination between food providers and Distributors (NGO's /volunteers). The system aims to ensure real-time redistribution by notifying nearby distributors (within a 5km radius) and enforcing transparency through mandatory distribution proofs.

1.3. Scope:

In-Scope Features:

1. User registration and secure login for providers, Distributors, and admin.
2. Admin verification of legal documents for providers and Distributors.
3. Providers can post surplus food details (quantity, type, expiry time, location).
4. Distributors can view available food within their area.
5. Real-time notifications to distributors within 5km.
6. Distributors can claim available food.
7. Providers confirm pickup requests.
8. Pickup confirmation and transaction record storage.
9. Distributors upload distribution proof (photo or report).
10. Monitoring system with warning/blocking for misuse.
11. Admin dashboard for management, tracking, and report generation.

Out-of- Scope Features:

1. Mobile application version.
2. The system doesn't include transportation service or delivery tracking.
3. No financial transactions are included in this system.

2. Design and Implementation Constraints

2.1. Programming Language:

Frontend: HTML, CSS, React, Tailwind CSS, JavaScript will be used for responsive, component-based UI, allowing for dynamic updates.

Backend: Node.js and Express.js will be used to handle asynchronous requests, which is critical for the notification system.

2.2. Database and Servers:

Database: MongoDB will be used.

Authentication: JWT for secure, stateless user sessions.

3. User Classes and Characteristics

- **Food Providers (Restaurants, Hotels, Community Centers, Households):** Post surplus food description (quantity, expiry), confirm or reject a pickup request, confirm pickup, view distribution history with proofs.
- **Food distributors (Distributors, Volunteers):** Get verified via legal documents, get alert for food updates within 5km, request for food pickup, get notification for pickup approval, confirm pickup, upload distribution proof.
- **System Admin (Super User):** Verify user authentication via legal documents, monitor distribution proofs, block users with two missing distribution proofs.

4. Functional Requirements (FR)

ID	Requirement Name	Description	Users	Priority
FR-1	Registration	The system shall allow users (both providers and distributors) to create account.	Providers, distributors	High
FR-2	Submit documents	Users (both providers and distributors) shall submit their legal documents for authentication and verification.	Providers, distributors	High
FR-3	Login	The system shall allow users (both providers, distributors and admin) to login to their accounts.	Providers, distributors, admin	High
FR-4	Role based-Profile/ dashboard	The system shall provide role based separate portals/ dashboards for users to post foods, view history, update food status and request for food, confirm pickup status and so on.	Providers, distributors	High
FR-5	Post food	Providers shall post surplus food descriptions and details.	Providers	High
FR-6	Get request notification	Providers shall get notification when any distributor requests for food pickup.	Providers	High
FR-7	View distributor history	Providers shall be able to view distributor's transaction history and distribution proofs.	Providers	High
FR-8	Accept or reject request	Providers shall accept or reject pickup requests based on distributor's history and proofs.	Providers	High
FR-9	Confirm pickup	Both providers and distributors shall confirm and update status when any food is picked up.	Providers, distributors	High

FR-10	Logout	The system shall allow users to logout from their accounts.	Providers, distributors, admin	High
FR-11	Get nearby food notification	Distributors shall get a notification if any food is posted in 5km range.	Distributors	High
FR-12	View food	Distributors shall be able to view food list and food descriptions.	Distributors	High
FR-13	Request for food	Distributors shall be able to request for food that are posted by providers.	Distributors	High
FR-14	Get request approval or rejection notification	Distributors shall get a notification when provider approves or rejects his request.	Distributors	High
FR-15	Upload proofs	Distributors shall upload food distribution details and proofs (photos, videos).	Distributors	High
FR-16	Verify user documents	Admins shall verify user documents to authenticate their account for registration.	Admins	High
FR-17	Admin dashboard	Admins shall be provided a dashboard for user activity monitoring and handling.	Admins	High
FR-18	Monitor proofs	Admins shall monitor distributor's food distribution proofs.	Admins	Medium
FR-19	Block users	Admin will block distributor when his maximum two distribution proofs are missing.	Admins	High
FR-20	Send notifications	System shall send notification to distributor when any food is posted in 5km range and also food request approval or rejection.	System	High

5. Non-Functional Requirements

5.1. Performance:

- User actions should be loaded within 2seconds.
- Food listing and search results shall load within 3 seconds.
- Notification delivery to nearby distributors shall occur within 10 seconds of food posting.

5.2. Security:

- Secured authentication is a must for users.
- Passwords shall be stored using secure hashing techniques.
- Sensitive data (documents, contact details) shall be protected from unauthorized access.
- The system shall prevent unauthorized actions through role-based access control (admin, provider, distributor).

5.3. Usability:

- Simple and user-friendly interface.

5.4. Availability:

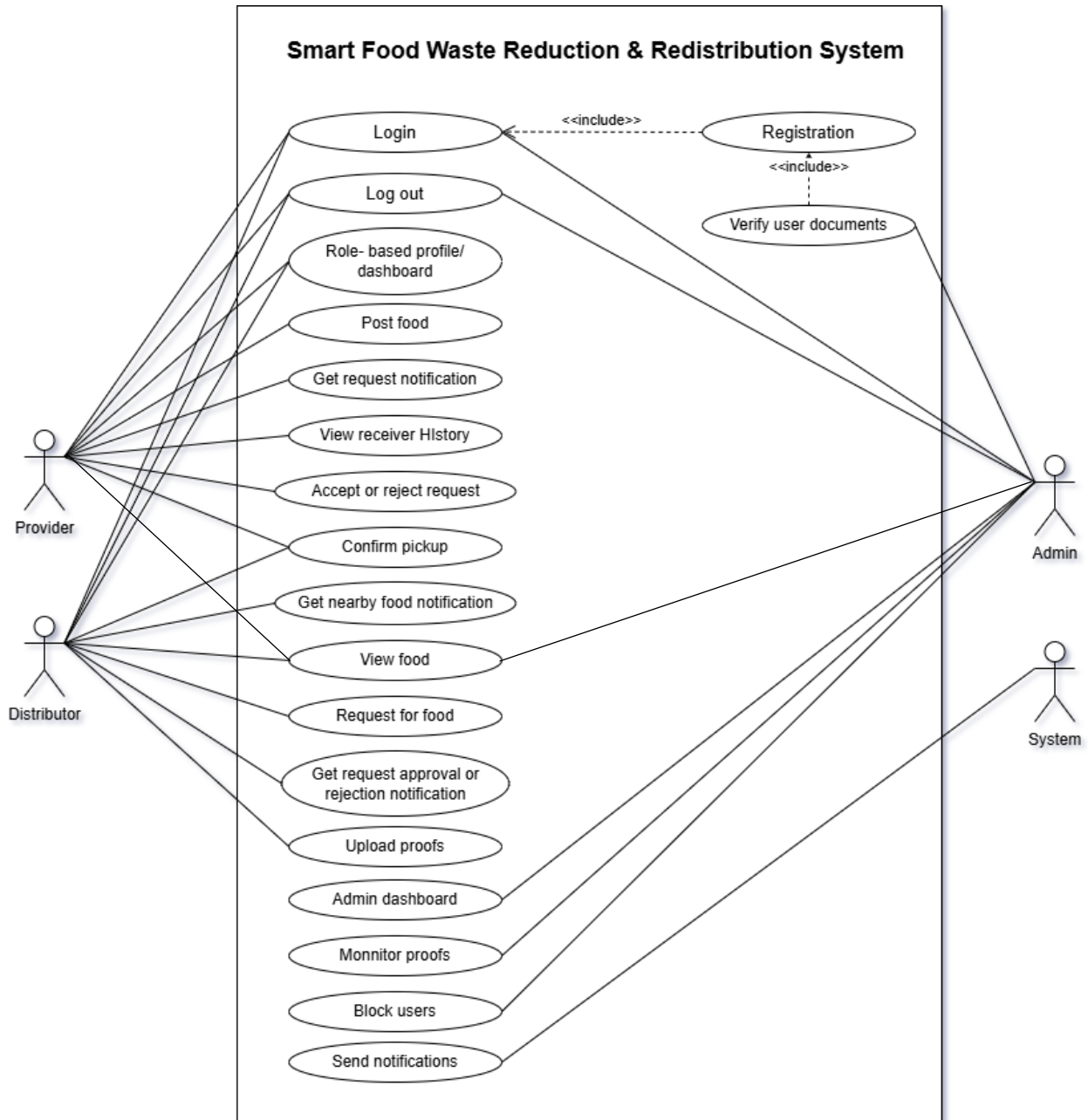
- The system shall be available 24/7 except during scheduled maintenance.
- System downtime for maintenance shall not exceed 2 hours per month.

5.5. Scalability:

- The system shall support at least 1,000 concurrent users without performance degradation.

6. System Design Diagrams

6.1. Use Case Diagram



6.2. Use Case Descriptions (High priority features)

UC-01: Submit Documents

Actors: Provider, Distributor

Description: Users submit legal documents for verification.

Preconditions: User has initiated registration process.

Main Flow:

1. User uploads legal documents.
2. System stores documents.
3. Document status set to "Pending Verification".

Postcondition: Documents are available for admin review.

UC-02: Post Food

Actors: Provider

Description: Provider posts surplus food details.

Preconditions: Provider must be verified and logged in.

Main Flow:

1. Provider enters food details (type, quantity, location, expiry time).
2. Provider submits form.
3. System stores food post.
4. System sends notification to nearby distributors.

Postcondition: Food becomes available for claim.

UC-03: View Food

Actors: Distributor

Description: Distributor views available food posts.

Preconditions: Distributor must be logged in.

Main Flow:

1. Distributor accesses food list.
2. System displays available food sorted by expiry and distance.

Postcondition: Distributor can select food to claim.

UC-04: Claim Food

Actors: Distributor

Description: Distributor requests to collect posted food.

Preconditions: Food must be available.

Main Flow:

1. Distributor selects food post.
2. Distributor clicks claim.
3. System records claim request.
4. System notifies provider.

Postcondition: Claim status set to "Pending".

UC-05: Approve/Reject Claim

Actors: Provider

Description: Provider reviews and decides on claim request.

Preconditions: Claim must be pending.

Main Flow:

1. Provider views claim request.
2. Provider checks distributor history.
3. Provider approves or rejects request.
4. System notifies distributor.

Postcondition: Claim status updated.

UC-06: Confirm Pickup

Actors: Provider, Distributor

Description: Confirms that food has been collected.

Preconditions: Claim must be approved.

Main Flow:

1. Distributor confirms pickup.
2. Provider confirms pickup.
3. System updates pickup status to completed.

Postcondition: Pickup process completed.

UC-07: Upload Distribution Proof

Actors: Distributor

Description: Distributor uploads proof after distributing food.

Preconditions: Pickup must be completed.

Main Flow:

1. Distributor uploads photo/report.
2. System stores proof.
3. Proof status set to "Pending Review".

Postcondition: Proof submitted for admin monitoring.

UC-08: Upload Distribution Proof

Actors: Distributor

Description: Distributor uploads proof after distributing food.

Preconditions: Pickup must be completed.

Main Flow:

1. Distributor uploads photo/report.
2. System stores proof.
3. Proof status set to "Pending Review".

Postcondition: Proof submitted for admin monitoring.

UC-9: Monitor Distribution Proof

Actors: Admin

Description: Admin reviews uploaded distribution proofs.

Preconditions: Proof must be uploaded.

Main Flow:

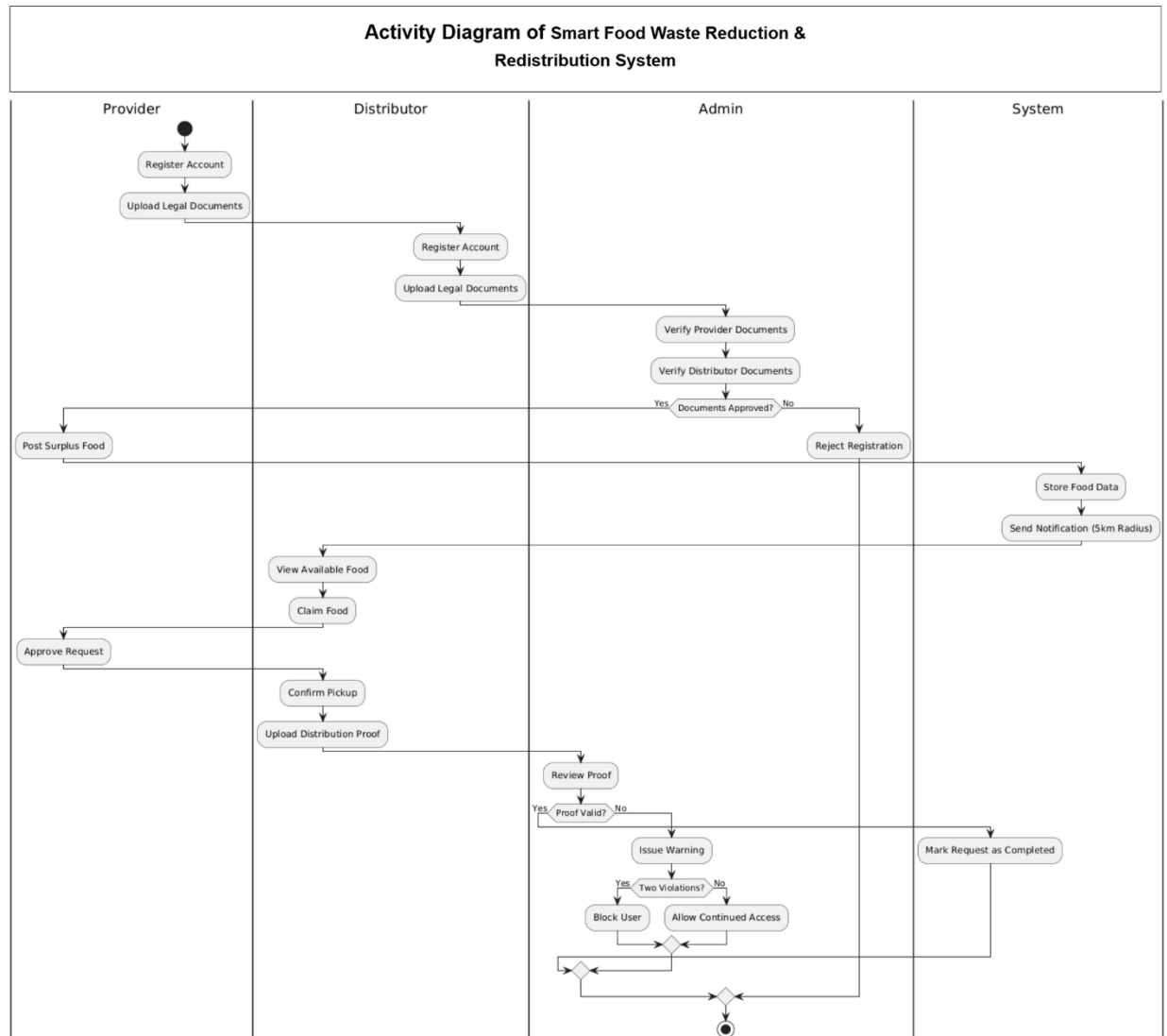
1. Admin reviews proof details.
2. Admin marks proof as valid or rejected.

Alternate Flow:

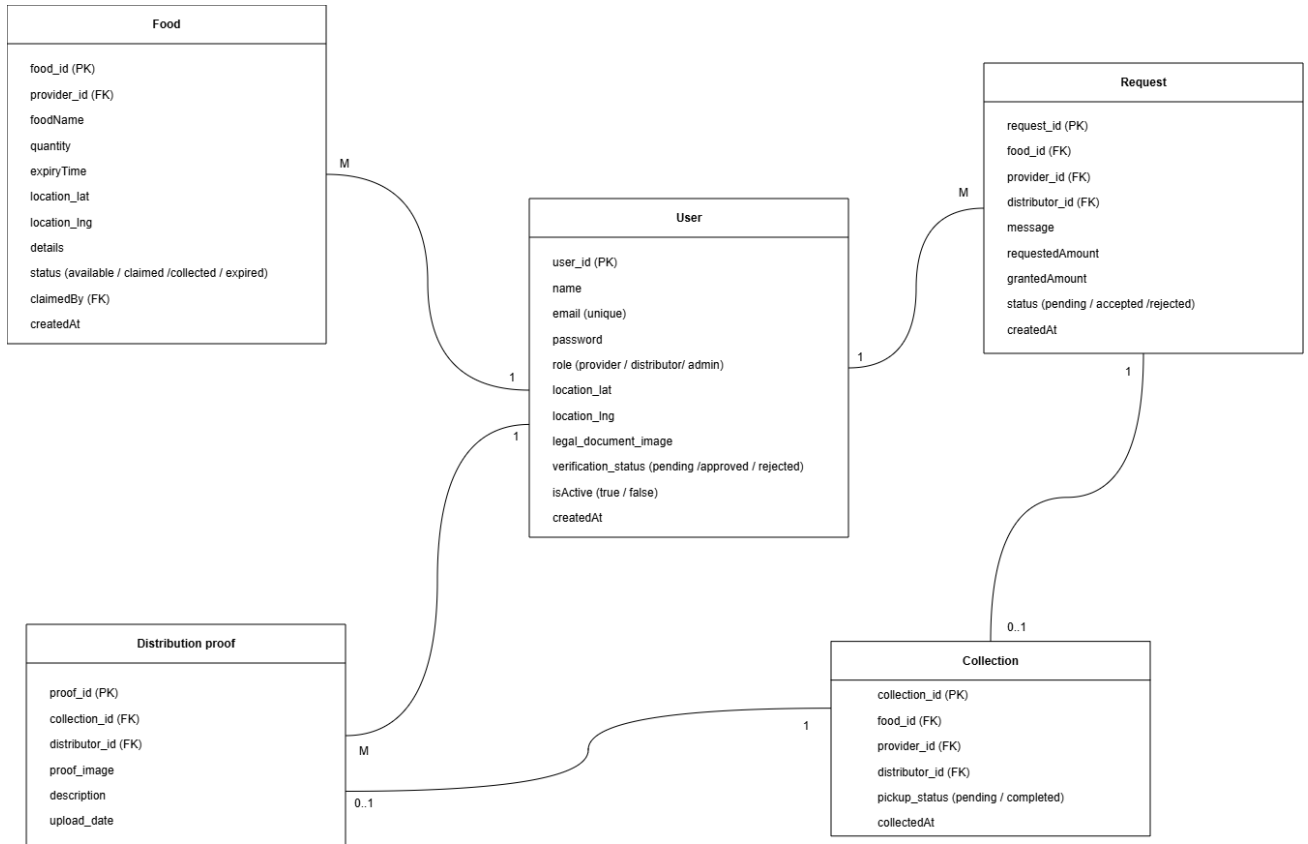
- If fraud detected → Admin blocks distributor.

Postcondition: Proof status updated.

6.3. Activity Diagram (An insight of overall system activity flow)



6.4. Entity Relationship Diagram (ERD) (Based on Database)



7. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibility
Tanvir Ahmed	0242310005341015	UI and Frontend: Interface design, SRS Documentation, React components, State management.
Md Mahbub Alam	0242310005341219	Testing & Documentation: Testing, SRS documentation, System Admin features verification.
Md Muhtasim Fuad Rahat	0242310005341171	Backend Development & Project Deployment: API build and integrate with frontend, Geo-spatial logic implementation (ModistributorsDB), Authentication system, Map integration, deploy frontend and backend